

Estimating Output Gap with the Time-Varying Slope New Keynesian Phillips Curve

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Abstract

We introduce a multivariate unobserved components model within the New Keynesian Phillips Curve framework that jointly incorporate a time-varying slope and correlated shocks to estimate output gap. While various model specifications have been proposed in the literature, none have jointly incorporate these features and systematically tested which feature is appropriate. To address this gap, we develop Bayesian model comparison method to evaluate the adequacy of various model specifications. Applying the proposed model to U.S. quarterly data from 1960Q1 to 2024Q2, we find that the output gap estimates yields larger amplitudes and greater persistence than those from models assuming constant slopes or uncorrelated shocks. These gaps are negatively correlated with potential output and positively correlated with inflation shocks. We also find a negative correlation between potential output and inflation shocks. In addition, the Bayesian model comparison exercise suggests that the proposed model significantly outperforms competing models, underscoring the importance of jointly considering these features.

Keywords: Output gap; New Keynesian Phillips curve; Multivariate correlated unobserved components model; Time-varying slope; Bayesian model comparison.

JEL codes: C11, C52, E31, E32

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1 Introduction

Estimating output gap, defined as the difference between actual output and potential output, is important for both academics and policymakers. Output can be decomposed into long-term trend and short-term cyclical components. Cyclical fluctuations influence inflation through the Phillips Curve relationship, while inflation movements provide information on economic slack (the output gap). Consequently, a vast literature exploits the Phillips curve to estimate the output gap, with multivariate unobserved components models serving as a standard tool for this purpose.

Conventionally, multivariate unobserved components models that include the Phillips curve impose two restrictions: (1) trend and cycle shocks are independent, and (2) the inflation–output gap relationship is constant (e.g., [Kuttner, 1994](#); [Roberts, 2001](#)). The first assumption may overlook interactions suggested by economic theory, such as the time-to-build effect ([Kydland and Prescott, 1982](#)), and univariate evidence questions this independence (e.g., [Morley et al., 2003](#); [Dungey et al., 2015](#); [Grant and Chan, 2017a,b](#)). The second assumption of a constant Phillips curve also faces challenges. Extensive literature documents structural instability in macroeconomic relationships (see, e.g., [Canova, 1993](#); [Cogley and Sargent, 2001](#); [Primiceri, 2005](#); [Koop and Potter, 2007](#)), and many studies show the Phillips curve slope changes over time or even has flattened (e.g., [Ball and Mazumder, 2011](#); [Stella and Stock, 2013](#); [Coibion and Gorodnichenko, 2015](#); [Chan et al., 2016](#); [Stock and Watson, 2020](#); [Ball and Mazumder, 2019](#); [Fu, 2020](#); [Del Negro et al., 2020](#); [Inoue et al., 2024](#)). This raises questions about whether inflation remains useful for estimating the output gap. Surprisingly, few studies jointly model correlated shocks and a potentially evolving Phillips curve slope for output gap estimation.

Our main contribution in this paper is to develop a new multivariate correlated unobserved components model that accommodates correlated trend-cycle shocks and a time-varying inflation–output gap relationship, along with an efficient MCMC estimation scheme. We build upon the multivariate correlated unobserved components model introduced by [Basistha and Nelson \(2007\)](#), which uses the new Keynesian Phillips curve to

estimate the output gap without imposing restrictive independence among shocks. Subsequent extensions adapt this baseline model to various settings, such as incorporating country-specific differences ([Gonzalez-Astudillo, 2019](#)) or introducing adaptive learning-based inflation expectations ([Panovska and Ramamurthy, 2022](#)), yet they all assume a constant Phillips curve slope. In contrast, some studies develop multivariate time-varying parameter multivariate unobserved components models that allow the Phillips curve slope to evolve over time (e.g., [Stella and Stock, 2013](#); [Chan et al., 2016](#); [Berger et al., 2016](#); [Fu, 2020](#); [Kabundi et al., 2023](#)), but do not account for correlated trend and cycle shocks. Our proposed time-varying parameter multivariate correlated unobserved components (TVP-MCUC) model addresses both issues by capturing the time-varying Phillips curve slope and correlated shocks, enabling us to utilize the evolving Phillips curve to estimate the output gap even in the presence of correlated trend and cycle components.

In this paper, we introduce a new MCMC method for estimating our proposed TVP-MCUC model. This method exploits the joint Gaussian structure of the measurement and state equations within the multivariate unobserved components framework, which simplifies implementation when the innovations are jointly correlated. Another key feature of our MCMC method is that it builds upon the band and sparse algorithm developed in [Chan and Jeliazkov \(2009\)](#) and [McCausland \(2012\)](#), demonstrating significantly greater efficiency than conventional Kalman filter-based algorithms.

We also contribute to the literature by introducing a Bayesian model comparison scheme that jointly evaluates the adequacy of correlated shocks and time-varying Phillips curve slopes in multivariate unobserved components models. Previous research demonstrates that accounting for correlated trend-cycle shocks can significantly alter estimates (e.g., [Morley et al., 2003](#)). Parameter instability also poses policy concerns: if the Phillips curve has flattened, recent inflation surges may require more aggressive stabilization measures. Without formal testing, incorporating time-varying parameters and correlated shocks risks model misspecification, potentially misleading policymakers and researchers. While some studies assess whether correlated shocks are supported by the data (e.g., [Grant and Chan, 2017a,b](#)) and others examine the time variation in the Phillips curve slope

(e.g., [Berger et al., 2016](#); [Fu, 2020](#)), few employ Bayesian model comparison based on marginal likelihood to evaluate both specifications simultaneously. To address this gap, we develop importance sampling methods to compute the marginal likelihoods of multivariate unobserved components models with correlated shocks or time-varying Phillips curve slopes. More specifically, we adapt the marginal likelihood computation method of [Chan \(2023\)](#), originally designed for large VAR models with stochastic volatility, to the multivariate unobserved components model framework. Similar to the approaches in [Grant and Chan \(2017a,b\)](#) for univariate unobserved components models, our method delivers more efficient and accurate inference in a multivariate setting and subject to constraints.

We apply our methodology to estimate the U.S. output gap through the lens of the new Keynesian Phillips curve. First, we conduct a comprehensive model comparison to determine which features—correlated shocks or time-varying Phillips curve—are supported by the data. Our results show that incorporating both these features fits best. Second, using our proposed model, we find that output gaps are negatively correlated with potential output and positively correlated with inflation shocks, consistent with other multivariate correlated unobserved components studies ([Basistha and Nelson, 2007](#); [Panovska and Ramamurthy, 2022](#)). Unlike these studies, we also discover that potential output is negatively correlated with inflation shocks. Our estimated output gap features larger amplitudes than competing models, particularly during the Great Recession and the COVID-19 pandemic, suggesting that ignoring a time-varying inflation-output gap relationship and correlated shocks within a unified multivariate unobserved components framework may lead to an underestimation of the variability in the output gap. Moreover, our model captures significant negative output gaps even during normal periods, attributed to sudden upward shifts in the long-term output path caused by negative inflation shocks. Actual output temporarily falls below this new trend until it catches up, resulting in large negative output gaps even in stable times. Finally, we find that although the Phillips curve slope declined from the early 1980s to the mid-1990s, it remains robust enough for inflation to play an important role in output gap estimation.

The rest of this paper is organized as follows. Section 2 introduces the TVP-MCUC model. Section 3 details our Bayesian estimation scheme. Section 4 presents our Bayesian model comparison method. Section 5 discusses our empirical results, and Section 6 concludes.

2 The TVP-MCUC Model

We start from a new Keynesian Phillips curve equation similar to that of [Basistha and Nelson \(2007\)](#), but with a time-varying slope of the Phillips curve:

$$\pi_t = \tau_t + \delta_t g_t, \quad (1)$$

where π_t is the inflation rate, g_t is the output gap arising from nominal rigidities, and τ_t represents inflation expectations. These inflation expectations are derived from linear projections on observable factors, including survey expectations π_t^{se} and lagged inflation π_{t-1} (see equation (3)). Equation (1) corresponds to the well-known hybrid New Keynesian Phillips Curve (NKPC) in the spirit of [Galí and Gertler \(1999\)](#). To capture gradual changes over time in the relationship between inflation and the output gap, we assume that the slope of the Phillips curve, δ_t , follows a random walk process as stated in (2). This is a standard approach to modeling slope instability in the Phillips curve (e.g., [Ball and Mazumder, 2011](#); [Stella and Stock, 2013](#); [Chan et al., 2016](#); [Ball and Mazumder, 2019](#)):

$$\delta_t = \delta_{t-1} + \varepsilon_{\delta,t}, \quad \varepsilon_{\delta,t} \sim \mathcal{N}(0, \sigma_\delta^2) \quad (2)$$

with the initial state $\delta_0 \sim \mathcal{N}(\delta_{00}, V_{\delta_0})$.

Following [Basistha and Nelson \(2007\)](#), the state equation for inflation expectations τ_t is specified as:

$$\tau_t = \beta_0 + \beta_1 \pi_t^{se} + (1 - \beta_1) \pi_{t-1} + \varepsilon_{\tau,t}, \quad (3)$$

where $\varepsilon_{\tau,t}$ captures both shocks to expected inflation and direct supply shocks to inflation. To ensure long-run neutrality, we impose the restriction $0 < \beta_1 < 1$.

We decompose the observed output z_t into a cyclical component g_t , representing the output gap, and a trend component r_t , representing potential output:

$$z_t = g_t + r_t. \quad (4)$$

The output gap g_t is assumed to follow a stationary AR(2) process, aiding in the identification of correlations among shocks (see, [Morley et al., 2003](#); [Trenkler and Weber, 2016](#)). Following [Morley et al. \(2003\)](#) and [Basistha and Nelson \(2007\)](#), the potential output r_t evolves as a random walk with a drift μ , which can be interpreted as the average growth rate of trend output.¹ Thus:

$$g_t = \phi_1 g_{t-1} + \phi_2 g_{t-2} + \varepsilon_{g,t}, \quad (5)$$

$$r_t = \mu + r_{t-1} + \varepsilon_{r,t}, \quad (6)$$

with $r_0 \sim \mathcal{N}(r_{00}, V_{r_0})$ and $\phi = (\phi_1, \phi_2)' \in \mathcal{R}$, where $\mathcal{R}$ denotes the stationarity region of the AR(2) process.

The variance-covariance matrix of the innovations $\varepsilon_t = (\varepsilon_{\tau,t}, \varepsilon_{g,t}, \varepsilon_{r,t})'$ in (3), (5), and (6) is given by:

$$\text{Var}(\varepsilon_t) = \Sigma = \begin{pmatrix} \sigma_{\tau}^2 & \sigma_{\tau g} & \sigma_{\tau r} \\ \sigma_{\tau g} & \sigma_g^2 & \sigma_{gr} \\ \sigma_{\tau r} & \sigma_{gr} & \sigma_r^2 \end{pmatrix}. \quad (7)$$

In summary, by allowing for correlated shocks and a time-varying slope of the Phillips curve, our proposed model extends the multivariate unobserved components framework

¹While evidence against a constant drift μ exists (e.g., [Perron and Wada, 2009](#); [Luo and Startz, 2014](#); [Grant and Chan, 2017a](#)), [Grant and Chan \(2017b\)](#) show that an unobserved correlated components model with constant drift outperforms most competing models allowing for drift instability. Only a model with a break in the drift at 2007 slightly outperforms the constant-drift specification. For simplicity, we maintain a constant drift in our specification.

to encompass special cases both with and without these features. In Section 5.2, we use this model structure in a Bayesian model comparison exercise to evaluate the adequacy of allowing for a time-varying slope of the Phillips curve and correlated trend-cycle shocks.

3 Bayesian Estimation

In this section we first detail the priors of the model parameters. We then outline the posterior simulator to estimate the model described in (1)–(7). We assume the following independent priors for model parameters $(\beta_0, \beta_1, \boldsymbol{\phi}, \mu, r_0, \delta_0, \boldsymbol{\Sigma}, \sigma_\delta^2)$:

$$\begin{aligned} \beta_0 &\sim \mathcal{N}(\beta_{00}, V_{\beta_{00}}), \quad \beta_1 \sim \mathcal{TN}_{\{\beta_1 \in (0,1)\}}(\beta_{10}, V_{\beta_{10}}), \quad \boldsymbol{\phi} \sim \mathcal{TN}_{\{\boldsymbol{\phi} \in \mathcal{R}\}}(\boldsymbol{\phi}_0, \mathbf{V}_{\boldsymbol{\phi}_0}), \\ \mu &\sim \mathcal{N}(\mu_0, V_{\mu_0}), \quad r_0 \sim \mathcal{N}(r_{00}, V_{r_0}), \quad \delta_0 \sim \mathcal{N}(\delta_{00}, V_{\delta_0}), \\ \boldsymbol{\Sigma} &\sim \mathcal{IW}(\mathbf{S}, \nu), \quad \sigma_\delta^2 \sim \mathcal{IG}(\alpha, \gamma), \end{aligned} \quad (8)$$

where $\mathbf{1}(\cdot)$ denote the indicator function and $\mathcal{TN}_A$ denotes a truncated Gaussian distribution with its support restricted on A . For $\beta_0, \beta_1, \boldsymbol{\phi}, \mu, r_0, \delta_0$, we assume relatively large prior variance with $V_{\beta_{00}} = 1, V_{\beta_{10}} = 1, \mathbf{V}_{\boldsymbol{\phi}} = I_2, V_{\mu_0} = 1, V_{r_0} = 10, V_{\delta_0} = 1$. For prior mean, we set $\beta_{00} = -0.2, \beta_{10} = 0.3, \boldsymbol{\phi} = (1.6; -0.7)', \mu_0 = 0.75, r_0 = 815, \delta_0 = 0.2$. For $\boldsymbol{\Sigma}$ and σ_δ^2 , we set $\mathbf{S} = \text{diag}(1, 1, 1), \nu = 3, \alpha = 20$, and $\gamma = 0.0005 \times (\alpha - 2)$.

To facilitate the discussion of the posterior simulator, let observed data $\mathbf{y} = (\boldsymbol{\pi}', \mathbf{z}')'$ where $\boldsymbol{\pi} = (\pi_1, \dots, \pi_T)'$ and $\mathbf{z} = (z_1, \dots, z_T)'$, $\boldsymbol{\theta} = (\beta_0, \beta_1, \boldsymbol{\phi}', \mu, r_0, \delta_0)'$, $\mathbf{r} = (r_1, \dots, r_T)'$ and $\boldsymbol{\delta} = (\delta_1, \dots, \delta_T)'$. Posterior draws can be obtained by sequentially drawing from the following conditional posteriors:

1. $p(\mathbf{r}|\boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma}, \mathbf{y})$;
2. $p(\boldsymbol{\delta}|\mathbf{r}, \boldsymbol{\theta}, \boldsymbol{\Sigma}, \sigma_\delta^2, \mathbf{y})$;
3. $p(\boldsymbol{\theta}|\mathbf{r}, \boldsymbol{\delta}, \boldsymbol{\Sigma}, \mathbf{y})$;
4. $p(\boldsymbol{\Sigma}|\mathbf{r}, \boldsymbol{\delta}, \boldsymbol{\theta}, \mathbf{y})$;

5. $p(\sigma_\delta^2 | \boldsymbol{\delta}, \mathbf{y})$.

Step 3 to Step 5 involves standard sampling techniques, and therefore we leave the details to Appendix A. We discuss Step 1 and Step 2 in this section.

Step 1. Here we provide a straightforward approach to obtain the conditional posterior of $\mathbf{r}$. Our approach is particularly advantageous in our framework where the conditional likelihood function of $\mathbf{r}$ is complex due to the presence of correlated errors in both the observation and state equations.

Our approach for finding the conditional posterior of $\mathbf{r}$ is based on the fact that the observable and the state parameters in our model are jointly multivariate Gaussian. Specifically, we first substitute equations (3) and (4) into equations (1), and (4) to (5). The model (1)-(6) can be represented as follows:

$$\pi_t - \delta_t z_t + \delta_t r_t = \beta_0 + \beta_1 \pi_t^{se} + (1 - \beta_1) \pi_{t-1} + \varepsilon_{\tau,t}, \quad (9)$$

$$z_t - r_t = \phi_1(z_{t-1} - r_{t-1}) + \phi_2(z_{t-2} - r_{t-2}) + \varepsilon_{g,t}, \quad (10)$$

$$r_t = \mu + r_{t-1} + \varepsilon_{r,t}. \quad (11)$$

Define $\tilde{\mathbf{x}}_t = (\pi_t, z_t, r_t)'$, we can rewrite equations (9) - (11) more compactly as

$$\mathbf{A}_{0,t} \tilde{\mathbf{x}}_t = \mathbf{c}_t + \mathbf{A}_1 \tilde{\mathbf{x}}_{t-1} + \mathbf{A}_2 \tilde{\mathbf{x}}_{t-2} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \quad (12)$$

where $\mathbf{c}_t = (\beta_0 + \beta_1 \pi_t^{se}, 0, \mu_t)'$ with $\mu_1 = \mu + r_0$, $\mu_t = \mu$ for $t = 2, \dots, T$, and

$$\mathbf{A}_{0,t} = \begin{pmatrix} 1 & -\delta_t & \delta_t \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{A}_1 = \begin{pmatrix} 1 - \beta_1 & 0 & 0 \\ 0 & \phi_1 & -\phi_1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{A}_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \phi_2 & -\phi_2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Therefore by stacking equation (12) over $t = 1, \dots, T$, we have

$$\mathbf{A} \tilde{\mathbf{x}} = \mathbf{c} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_T \otimes \boldsymbol{\Sigma}), \quad (13)$$

where $\tilde{\mathbf{x}} = (\tilde{\mathbf{x}}'_1, \dots, \tilde{\mathbf{x}}'_T)'$, $\mathbf{c} = (\mathbf{c}'_1, \dots, \mathbf{c}'_T)'$, $\boldsymbol{\varepsilon} = (\boldsymbol{\varepsilon}'_1, \dots, \boldsymbol{\varepsilon}'_T)'$ and

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{0,1} & \mathbf{0} & \mathbf{0} & \cdots & \cdots & \cdots & \mathbf{0} \\ -\mathbf{A}_1 & \mathbf{A}_{0,2} & \mathbf{0} & \ddots & \ddots & \ddots & \vdots \\ -\mathbf{A}_2 & -\mathbf{A}_1 & \mathbf{A}_{0,3} & \mathbf{0} & \ddots & \ddots & \vdots \\ \mathbf{0} & -\mathbf{A}_2 & -\mathbf{A}_1 & \mathbf{A}_{0,4} & \mathbf{0} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \mathbf{0} \\ \vdots & \ddots & \mathbf{0} & -\mathbf{A}_2 & -\mathbf{A}_1 & \mathbf{A}_{0,T-1} & \mathbf{0} \\ \mathbf{0} & \cdots & \cdots & \mathbf{0} & -\mathbf{A}_2 & -\mathbf{A}_1 & \mathbf{A}_{0,T} \end{pmatrix}$$

Equation (13) implies that $(\tilde{\mathbf{x}}|\boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma}) \sim \mathcal{N}(\boldsymbol{\mu}_{\tilde{\mathbf{x}}}, \mathbf{K}_{\tilde{\mathbf{x}}}^{-1})$, where $\boldsymbol{\mu}_{\tilde{\mathbf{x}}} = \mathbf{A}^{-1}\mathbf{c}$ and $\mathbf{K}_{\tilde{\mathbf{x}}} = \mathbf{A}'(\mathbf{I}_T \otimes \boldsymbol{\Sigma}^{-1})\mathbf{A}$.² To derive the conditional posterior of $\mathbf{r}$, we introduce a $3T \times 3T$ selection matrix such that $\mathbf{S}_r \tilde{\mathbf{x}} = \mathbf{x}$ with $\mathbf{x} = (\mathbf{r}', \mathbf{y}')'$. Notice that this transformation rearranges the elements in $\tilde{\mathbf{x}}$, resulting in the parameter vector $\mathbf{r}$ being ordered as the first block within $\mathbf{x}$. Since $\tilde{\mathbf{x}}$ is an Gaussian given $(\boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma})$, so is also the conditional distribution for $\mathbf{x}$:

$$(\mathbf{x}|\boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma}) \sim \mathcal{N}(\boldsymbol{\mu}_x, \mathbf{K}_x^{-1}), \quad (14)$$

where $\boldsymbol{\mu}_x = \mathbf{S}_r \boldsymbol{\mu}_{\tilde{\mathbf{x}}}$ and $\mathbf{K}_x = \mathbf{S}_r \mathbf{K}_{\tilde{\mathbf{x}}} \mathbf{S}_r'$. Here we use the fact that $\mathbf{S}_r^{-1} = \mathbf{S}_r'$. Next, we partition the mean $\boldsymbol{\mu}_x$ and the precision matrix $\mathbf{K}_x$ as

$$\boldsymbol{\mu}_x = \begin{pmatrix} \boldsymbol{\mu}_r \\ \boldsymbol{\mu}_y \end{pmatrix}, \quad \mathbf{K}_x = \begin{pmatrix} \mathbf{K}_r & \mathbf{K}_{r,y} \\ \mathbf{K}_{r,y}' & \mathbf{K}_y \end{pmatrix}$$

where $\boldsymbol{\mu}_r$ is $T \times 1$, $\boldsymbol{\mu}_y$ is $2T \times 1$, $\mathbf{K}_r$ is $T \times T$, $\mathbf{K}_y$ is $2T \times 2T$ and $\mathbf{K}_{r,y}$ is $T \times 2T$. Using the standard property of Gaussian distribution, the conditional posterior of $\mathbf{r}$ is given by

$$(\mathbf{r}|\boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma}, \mathbf{y}) \sim \mathcal{N}(\hat{\boldsymbol{\mu}}_r, \mathbf{K}_r^{-1}), \quad (15)$$

where $\hat{\boldsymbol{\mu}}_r = \boldsymbol{\mu}_r - \mathbf{K}_r^{-1} \mathbf{K}_{r,y} (\mathbf{y} - \boldsymbol{\mu}_y)$.

²Here we assume $\pi_t = 0$ and $z_t - r_t = 0$ if $t < 0$.

Step 2. We first let $\tilde{\mathbf{w}}_t = (\pi_t, g_t, r_t, \delta_t)'$ where $g_t = z_t - r_t$. It is straightforward to check that equations (9), (5), (6) and (2) can be rewritten as

$$\mathbf{B}_{0,t}\tilde{\mathbf{w}}_t = \mathbf{c}_{\tilde{w},t} + \mathbf{B}_1\tilde{\mathbf{w}}_{t-1} + \mathbf{B}_2\tilde{\mathbf{w}}_{t-2} + \mathbf{u}_t, \quad \mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{\Omega}), \quad (16)$$

where $\mathbf{u}_t = (\boldsymbol{\varepsilon}'_t, \varepsilon_{\delta,t})'$, $\mathbf{\Omega} = \text{diag}(\boldsymbol{\Sigma}, \sigma_\delta^2)$, $\mathbf{c}_{w,1} = (\beta_0 + \beta_1\pi_1^{se}, 0, \mu + r_0, \delta_0)'$, $\mathbf{c}_{w,t} = (\beta_0 + \beta_1\pi_t^{se}, 0, \mu, 0)'$ for $t = 2, \dots, T$, and

$$\mathbf{B}_{0,t} = \begin{pmatrix} 1 & 0 & 0 & -g_t \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{B}_1 = \begin{pmatrix} 1 - \beta_1 & 0 & 0 & 0 \\ 0 & \phi_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{B}_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \phi_2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Stacking equation (16) over $t = 1, \dots, T$, we have

$$\mathbf{B}\tilde{\mathbf{w}} = \mathbf{c}_{\tilde{w}} + \mathbf{u}, \quad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_T \otimes \mathbf{\Omega}), \quad (17)$$

where $\tilde{\mathbf{w}} = (\tilde{\mathbf{w}}'_1, \dots, \tilde{\mathbf{w}}'_T)'$, $\mathbf{c}_{\tilde{w}} = (\mathbf{c}'_{\tilde{w},1}, \dots, \mathbf{c}'_{\tilde{w},T})'$, $\mathbf{u} = (\mathbf{u}'_1, \dots, \mathbf{u}'_T)'$ and

$$\mathbf{B} = \begin{pmatrix} \mathbf{B}_{0,1} & \mathbf{0} & \mathbf{0} & \cdots & \cdots & \cdots & \mathbf{0} \\ -\mathbf{B}_1 & \mathbf{B}_{0,2} & \mathbf{0} & \ddots & \ddots & \ddots & \vdots \\ -\mathbf{B}_2 & -\mathbf{B}_1 & \mathbf{B}_{0,3} & \mathbf{0} & \ddots & \ddots & \vdots \\ \mathbf{0} & -\mathbf{B}_2 & -\mathbf{B}_1 & \mathbf{B}_{0,4} & \mathbf{0} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \mathbf{0} & -\mathbf{B}_2 & -\mathbf{B}_1 & \mathbf{B}_{0,T-1} & \mathbf{0} \\ \mathbf{0} & \cdots & \cdots & \mathbf{0} & -\mathbf{B}_2 & -\mathbf{B}_1 & \mathbf{B}_{0,T} \end{pmatrix}.$$

Equation (17) implies that $(\tilde{\mathbf{w}}|\boldsymbol{\theta}, \boldsymbol{\Sigma}, \sigma_\delta^2) \sim \mathcal{N}(\boldsymbol{\mu}_{\tilde{w}}, \mathbf{K}_{\tilde{w}}^{-1})$, where $\boldsymbol{\mu}_{\tilde{w}} = \mathbf{B}^{-1}\mathbf{c}_{\tilde{w}}$ and $\mathbf{K}_{\tilde{w}} = \mathbf{B}'(\mathbf{I}_T \otimes \mathbf{\Omega}^{-1})\mathbf{B}$. Similar to the deviation of the conditional posterior of $\mathbf{r}$ in Step 1, we construct a $3T \times 3T$ selection matrix $\mathbf{S}_\delta$ to reorder the elements in $\tilde{\mathbf{w}}$ such that $\mathbf{S}_\delta\tilde{\mathbf{w}} = \mathbf{w}$

such that $\mathbf{w} = (\boldsymbol{\delta}', \mathbf{r}', \boldsymbol{\pi}', \mathbf{g}')'$, and we have

$$(\mathbf{w}|\boldsymbol{\theta}, \boldsymbol{\Sigma}, \sigma_\delta^2) \sim \mathcal{N}(\boldsymbol{\mu}_w, \mathbf{K}_w^{-1}), \quad (18)$$

where $\boldsymbol{\mu}_w = \mathbf{S}_\delta \boldsymbol{\mu}_{\tilde{w}}$ and $\mathbf{K}_w = \mathbf{S}_\delta \mathbf{K}_{\tilde{w}} \mathbf{S}_\delta'$. Partitioning the mean $\boldsymbol{\mu}_w$ and the precision matrix, $\mathbf{K}_w$ as

$$\boldsymbol{\mu}_w = \begin{pmatrix} \boldsymbol{\mu}_\delta \\ \boldsymbol{\mu}_{(r,\pi,g)} \end{pmatrix}, \quad \mathbf{K}_w = \begin{pmatrix} \mathbf{K}_\delta & \mathbf{K}_{\delta,(r,\pi,g)} \\ \mathbf{K}_{\delta,(r,\pi,g)}' & \mathbf{K}_{(r,\pi,g)} \end{pmatrix}$$

such that $\boldsymbol{\mu}_\delta$ is $T \times 1$, $\boldsymbol{\mu}_{(r,\pi,g)}$ is $2T \times 1$, $\mathbf{K}_\delta$ is $2T \times 2T$, $\mathbf{K}_{(r,\pi,g)}$ is $2T \times 2T$ and $\mathbf{K}_{\delta,(r,\pi,g)}$. Therefore, we can obtain the posterior distribution of $\boldsymbol{\delta}$:

$$(\boldsymbol{\delta}|\mathbf{r}, \boldsymbol{\theta}, \boldsymbol{\Sigma}, \sigma_\delta^2, \mathbf{y}) \sim \mathcal{N}(\hat{\boldsymbol{\mu}}_\delta, \mathbf{K}_\delta^{-1}), \quad (19)$$

where $\hat{\boldsymbol{\mu}}_\delta = \boldsymbol{\mu}_\delta - \mathbf{K}_\delta^{-1} \mathbf{K}_{\delta,(r,\pi,g)} (\mathbf{m}_{(r,\pi,g)} - \boldsymbol{\mu}_{(r,\pi,g)})$ with $\mathbf{m}_{(r,\pi,g)} = (\mathbf{r}', \boldsymbol{\pi}', \mathbf{g}')'$.

It is worth pointing out that the precision matrices $\mathbf{K}_r$ and $\mathbf{K}_\delta$ in the Gaussian conditional posteriors specified in (15) and (19) are banded and sparse. Therefore, the precision sampler proposed by Chan and Jeliazkov (2009) can be used to efficiently draw from the conditional posteriors for $\mathbf{r}$ and $\boldsymbol{\delta}$ to further speed up computation.

4 Model Comparison via the Marginal Likelihood

This section describes an adaptive importance sampling estimator for estimating the marginal likelihood of our proposed TVP-MCUC model. The proposed method is built upon the approach proposed by Chan (2023). The technical details for estimating the marginal likelihood are provided in Appendices B and C.

In general, the marginal likelihood for a given model is defined as:

$$p(\mathbf{y}) = \int p(\mathbf{y}|\boldsymbol{\eta})p(\boldsymbol{\eta})d\boldsymbol{\eta},$$

where $\boldsymbol{\eta}$ is a vector of model parameter, $\mathbf{y}$ is the observed data, $p(\mathbf{y}|\boldsymbol{\eta})$ and $p(\boldsymbol{\eta})$ are the likelihood function and the prior density of the model, respectively. The marginal likelihood can be interpreted as the marginal density evaluated at the observed data under the prior. A larger value of the marginal likelihood indicates better fitting of the model.

In this paper we adopt the approach proposed by [Chan \(2023\)](#) to estimate the marginal likelihood of the TVP-MCUC model, which combines the conditional Monte Carlo method and adaptive importance sampling for improving the accuracy of marginal likelihood estimation. The key idea of this approach is first to obtain an integrated likelihood of the model by integrating out as many parameters as possible first. Then the marginal likelihood of the model is computed using the integrated likelihood with an importance sampling estimator.³ The proposal distribution of the importance sampling estimator is constructed using the information from the entire support of the posterior distribution defined by the integrated likelihood of the parameters and their corresponding prior densities.

In our NKPC modeling framework, the observed data is given by $\mathbf{y} = (\boldsymbol{\pi}', \mathbf{z}')'$ and the model parameters are $\boldsymbol{\eta} = (\mathbf{r}, \boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma}, \sigma_\delta^2)$. We partition the model parameters vector as $\boldsymbol{\eta} = (\boldsymbol{\eta}_*, \boldsymbol{\eta}_{-*})$ with $\boldsymbol{\eta}_* = (\boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma})$ and $\boldsymbol{\eta}_{-*} = (\mathbf{r}, \sigma_\delta^2)$, then the marginal likelihood of our proposed NKPC model can be expressed as:

$$\begin{aligned}
p(\mathbf{y}) &= \int p(\mathbf{y}|\boldsymbol{\eta})p(\boldsymbol{\eta})d\boldsymbol{\eta} \\
&= \int p(\mathbf{y}|\mathbf{r}, \boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma}, \sigma_\delta^2)p(\mathbf{r}, \boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma}, \sigma_\delta^2)d(\mathbf{r}, \boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma}, \sigma_\delta^2) \\
&= \int p(\mathbf{y}|\mathbf{r}, \boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma})p(\mathbf{r}|\sigma_\delta^2)p(\boldsymbol{\delta}|\sigma_\delta^2)p(\sigma_\delta^2)p(\boldsymbol{\Sigma})p(\boldsymbol{\theta})d(\mathbf{r}, \boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma}, \sigma_\delta^2) \\
&= \int p(\mathbf{y}|\boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma})p(\boldsymbol{\delta}|\sigma_\delta^2)p(\boldsymbol{\Sigma})p(\boldsymbol{\theta})d(\boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma}) \\
&= \int p(\mathbf{y}|\boldsymbol{\eta}_*)p(\boldsymbol{\eta}_*)d(\boldsymbol{\eta}_*),
\end{aligned} \tag{20}$$

³Similar approaches of using integrated likelihood to compute marginal likelihood can also be found in [Grant and Chan \(2017a\)](#) and [Chan and Eisenstat \(2015\)](#).

where $p(\mathbf{y}|\boldsymbol{\eta}_*) = p(\mathbf{y}|\boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma}) = \int p(\mathbf{y}|\mathbf{r}, \boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma})p(\mathbf{r}|r_0, \boldsymbol{\Sigma})d\mathbf{r}$ is the integrated likelihood for $\boldsymbol{\eta}_*$ with the state variable $\mathbf{r}$ integrated out, $p(\boldsymbol{\eta}_*) = p(\boldsymbol{\delta}|\delta_0)p(\boldsymbol{\Sigma})p(\boldsymbol{\theta})$ is the implied prior for $\boldsymbol{\eta}_*$ where $p(\boldsymbol{\delta}|\delta_0) = \int p(\boldsymbol{\delta}|\delta_0, \sigma_\delta^2)p(\sigma_\delta^2)d\sigma_\delta^2$.

Note that in the expression (20) we work with the integrated likelihood $p(\mathbf{y}|\boldsymbol{\eta}_*)$ rather than the complete-data likelihood $p(\mathbf{y}|\boldsymbol{\eta}_*, \boldsymbol{\eta}_{-*})$. As shown by Chan (2023), working with the integrated likelihood can always improve the estimation accuracy of the marginal likelihood. This is simply due to the Rao-Blackwell theorem, which shows that $\text{Var}(p(\mathbf{y}|\boldsymbol{\eta}_*)) < \text{Var}(p(\mathbf{y}|\boldsymbol{\eta}))$. The details of the analytical expressions for $p(\mathbf{y}|\boldsymbol{\eta}_*)$ and $p(\boldsymbol{\eta}_*)$ are provided in the Appendix B.

An importance sampling estimator can be used to compute the marginal likelihood given in (20). Specifically, suppose that there is a density function of $\boldsymbol{\eta}_*$, denoted as $q(\boldsymbol{\eta}_*)$, and $q(\boldsymbol{\eta}_*) > 0$ whenever $p(\mathbf{y}|\boldsymbol{\eta}_*)p(\boldsymbol{\eta}_*) > 0$, then the marginal likelihood can be computed using the following importance sampling estimator:

$$\widehat{p(\mathbf{y})}_{IS} = \frac{1}{M} \sum_{m=1}^M \frac{p(\mathbf{y}|\boldsymbol{\eta}_*^{(m)})p(\boldsymbol{\eta}_*^{(m)})}{q(\boldsymbol{\eta}_*^{(m)})}, \quad (21)$$

where $\boldsymbol{\eta}_*^{(1)}, \dots, \boldsymbol{\eta}_*^{(M)}$ are independent draws obtained from the importance sampling distribution $q(\boldsymbol{\eta}_*)$.

The choice of the importance sampling density $q(\cdot)$ plays a crucial role for accurately estimating the marginal likelihood. It is well-known that if the posterior density is used as the importance sampling density, i.e., $q^*(\boldsymbol{\eta}) = p(\boldsymbol{\eta}_*|\mathbf{y}) = \frac{p(\mathbf{y}|\boldsymbol{\eta}_*)p(\boldsymbol{\eta}_*)}{p(\mathbf{y})}$, then the associated importance sampling estimator (21) will have zero variance because $\frac{p(\mathbf{y}|\boldsymbol{\eta}_*^{(m)})p(\boldsymbol{\eta}_*^{(m)})}{q(\boldsymbol{\eta}_*^{(m)})} = \frac{p(\mathbf{y}|\boldsymbol{\eta}_*^{(m)})p(\boldsymbol{\eta}_*^{(m)})}{p(\boldsymbol{\eta}_*^{(m)}|\mathbf{y})} = p(\mathbf{y})$. However, using the posterior density as the importance sampling density is infeasible in practice, because the normalizing constant of the posterior distribution, i.e., $p(\mathbf{y})$, is unknown and is the quantity that we would like to estimate. One feasible approach is to choose the importance sampling density from a certain family of distributions that approximates the posterior density the best. To be specific, we write the importance density as $q(\cdot) = q(\cdot; \mathbf{v})$ to emphasize that the importance sampling density belong to a family of distributions which is parameterized by parameters

$\mathbf{v}$ hereinafter. Our aim is to find a density $q(\cdot; \mathbf{v})$ that is closest to the posterior density $p(\cdot|\mathbf{y})$. The closeness between the importance sampling distribution $q(\cdot; \mathbf{v})$ and the posterior distribution $p(\cdot|\mathbf{y})$ is measured using the Kullback-Leibler divergence:

$$\text{KL}(p(\cdot|\mathbf{y}), q(\cdot; \mathbf{v})) = \int p(\boldsymbol{\eta}_*|\mathbf{y}) \log \frac{p(\boldsymbol{\eta}_*|\mathbf{y})}{q(\boldsymbol{\eta}_*; \mathbf{v})} d\boldsymbol{\eta}_*.$$

It is known that the Kullback-Leibler divergence is a positive measure, and is equal to zero when the $p(\cdot|\mathbf{y})$ and $q(\cdot; \mathbf{v})$ are identical. Choosing the parameter $\mathbf{v}$ to minimize the Kullback-Leibler divergence is equivalent to solving the following maximization problem:

$$\tilde{\mathbf{v}} = \operatorname{argmax}_{\mathbf{v}} \int p(\boldsymbol{\eta}_*|\mathbf{y}) \log q(\boldsymbol{\eta}_*; \mathbf{v}) d\boldsymbol{\eta}_*.$$

However, this optimization problem is often challenging to solve analytically. Here we solve its numerical counterpart:

$$\hat{\mathbf{v}} = \operatorname{argmax}_{\mathbf{v}} \frac{1}{M} \sum_{m=1}^M \log q(\boldsymbol{\eta}_*^{(m)}; \mathbf{v}), \quad (22)$$

where $\boldsymbol{\eta}_*^{(1)}, \dots, \boldsymbol{\eta}_*^{(M)}$ are draws from the posterior distribution $p(\boldsymbol{\eta}_*|\mathbf{y})$. In our application we consider a family of importance sampling density functions of the form $q(\boldsymbol{\eta}_*; \mathbf{v}) = q_{\bar{\boldsymbol{\delta}}}(\bar{\boldsymbol{\delta}}; \mathbf{v}_{\bar{\boldsymbol{\delta}}}) \times q_{\bar{\boldsymbol{\theta}}}(\bar{\boldsymbol{\theta}}; \mathbf{v}_{\bar{\boldsymbol{\theta}}}) \times q_{\Sigma}(\Sigma; \mathbf{v}_{\Sigma})$, where $\bar{\boldsymbol{\delta}} = (\delta_0, \boldsymbol{\delta}')'$, $\bar{\boldsymbol{\theta}} = (\beta_0, \beta_1, \boldsymbol{\phi}', \mu, r_0)'$ and $\mathbf{v} = (\mathbf{v}_{\bar{\boldsymbol{\delta}}}, \mathbf{v}_{\bar{\boldsymbol{\theta}}}, \mathbf{v}_{\Sigma})$. With this setting, the optimization problem (22) can be further decomposed into the following optimization problems:

$$\hat{\mathbf{v}}_{\bar{\boldsymbol{\delta}}} = \operatorname{argmax}_{\mathbf{v}_{\bar{\boldsymbol{\delta}}}} \frac{1}{M} \sum_{m=1}^M \log q_{\bar{\boldsymbol{\delta}}}(\bar{\boldsymbol{\delta}}^{(m)}; \mathbf{v}_{\bar{\boldsymbol{\delta}}}), \quad (23)$$

$$\hat{\mathbf{v}}_{\Sigma} = \operatorname{argmax}_{\mathbf{v}_{\Sigma}} \frac{1}{M} \sum_{m=1}^M \log q_{\Sigma}(\Sigma^{(m)}; \mathbf{v}_{\Sigma}), \quad (24)$$

$$\hat{\mathbf{v}}_{\bar{\boldsymbol{\theta}}} = \operatorname{argmax}_{\mathbf{v}_{\bar{\boldsymbol{\theta}}}} \frac{1}{M} \sum_{m=1}^M \log q_{\bar{\boldsymbol{\theta}}}(\bar{\boldsymbol{\theta}}^{(m)}; \mathbf{v}_{\bar{\boldsymbol{\theta}}}), \quad (25)$$

where $(\bar{\boldsymbol{\delta}}^{(1)}, \bar{\boldsymbol{\theta}}^{(1)}, \Sigma^{(1)}), \dots, (\bar{\boldsymbol{\delta}}^{(M)}, \bar{\boldsymbol{\theta}}^{(M)}, \Sigma^{(M)})$ are posterior draws from $p(\bar{\boldsymbol{\delta}}, \bar{\boldsymbol{\theta}}, \Sigma|\mathbf{y})$.

In our application $q_{\delta}(\cdot; \hat{\mathbf{v}}_{\delta})$ and $q_{\Sigma}(\cdot; \hat{\mathbf{v}}_{\Sigma})$ are chosen to be the densities of a Gaussian distribution and an inverse-Wishart distribution, respectively, and $q_{\bar{\theta}}(\cdot; \hat{\mathbf{v}}_{\bar{\theta}})$ is chosen to be the density of a truncated Gaussian distribution with its support restricted to satisfy $\beta_1 \in (0, 1)$ and $\phi \in \mathcal{R}$. In Appendix C we provide details about the procedure of finding the optimal parameters $\hat{\mathbf{v}}_{\delta}$, $\hat{\mathbf{v}}_{\bar{\theta}}$ and $\hat{\mathbf{v}}_{\Sigma}$. Hence, given $\hat{\mathbf{v}} = (\hat{\mathbf{v}}_{\delta}, \hat{\mathbf{v}}_{\bar{\theta}}, \hat{\mathbf{v}}_{\Sigma})$, the marginal likelihood can be estimated using the following adaptive importance sampling estimator:

$$\widehat{p(\mathbf{y})}_{AIS} = \frac{1}{M} \sum_{m=1}^M \frac{p(\mathbf{y}|\boldsymbol{\eta}_*^{(m)})p(\boldsymbol{\eta}_*^{(m)})}{q(\boldsymbol{\eta}_*^{(m)}; \hat{\mathbf{v}})} \quad (26)$$

with $\boldsymbol{\eta}_*^{(1)}, \dots, \boldsymbol{\eta}_*^{(M)}$ being independent draws from $q(\boldsymbol{\eta}_*; \hat{\mathbf{v}})$.

However, a practical difficulty arises when using the estimator (26) to compute the marginal likelihood in the context of our proposed NKPC model: the prior density $p(\boldsymbol{\eta}_*)$ and the importance sampling density $q(\boldsymbol{\eta}_*; \hat{\mathbf{v}})$ do not have analytical expressions. This is because the autoregressive coefficients ϕ contained in $\boldsymbol{\eta}_*$ are restricted to $\phi \in \mathcal{R}$, which results in the exact normalizing constant of both the prior density $p(\boldsymbol{\eta}_*)$ and the proposed importance sampling density $q(\boldsymbol{\theta}_*; \hat{\mathbf{v}})$ being unknown. To address this issue, we incorporate a self-normalized scheme to automatically take into account the unknown normalizing constant when computing the marginal likelihood. For expository purposes, we write $p(\boldsymbol{\eta}_*) = c_p \tilde{p}(\boldsymbol{\eta}_*)$ and $q(\boldsymbol{\theta}_*; \hat{\mathbf{v}}) = c_q \tilde{q}(\boldsymbol{\theta}_*; \hat{\mathbf{v}})$ where $\tilde{p}(\boldsymbol{\eta}_*)$ and $\tilde{q}(\boldsymbol{\theta}_*; \hat{\mathbf{v}})$ are the known kernels for the prior density and importance sampling density respectively, and c_p and c_q are their corresponding unknown normalizing constant. We propose to use the following self-normalizing adaptive importance sampling estimator to compute the marginal likelihood of our proposed TVP-MCUC model:

$$\widehat{p(\mathbf{y})}_{SAIS} = \frac{\frac{1}{M} \sum_{m=1}^M W_m p(\mathbf{y}|\boldsymbol{\eta}_*^{(m)})}{\frac{1}{M} \sum_{m=1}^M W_m}, \quad (27)$$

where $W_m = \frac{\tilde{p}(\boldsymbol{\eta}_*^{(m)})}{\tilde{q}(\boldsymbol{\eta}_*^{(m)}; \hat{\mathbf{v}})}$, and $\boldsymbol{\eta}_*^{(1)}, \dots, \boldsymbol{\eta}_*^{(M)}$ are independent draws from $q(\boldsymbol{\eta}_*; \hat{\mathbf{v}})$. It is straightforward to check that the estimator (27) is simulation-consistent for estimating the marginal likelihood because $\frac{1}{M} \sum_{m=1}^M W_m p(\mathbf{y}|\boldsymbol{\eta}_*^{(m)}) \rightarrow \frac{c_q}{c_p} p(\mathbf{y})$ and $\frac{1}{M} \sum_{m=1}^M W_m \rightarrow \frac{c_q}{c_p}$.

The details regarding the analytical expressions for $\tilde{p}(\boldsymbol{\eta}_*)$ and $\tilde{q}(\boldsymbol{\eta}_*; \hat{\mathbf{v}})$ are provided in Appendices B and C.

5 Empirical Results

In this section, we first describe the dataset used for our empirical application in Section 5.1. We then conduct a formal model comparison exercise in Section 5.2 to assess our proposed model against other competing models, highlighting the importance of allowing for a time-varying slope of the Phillips Curve and correlated trend-cycle shocks. Lastly, the estimation results and the output gap estimates of our proposed TVP-MCUC model are reported in Sections 5.3 and 5.4.

5.1 Data

Our data consists of quarterly U.S. series from 1960Q1 to 2024Q2, including the urban Consumer Price Index (CPI), Real Gross Domestic Product (GDP), and a survey measure of inflationary expectations. The data are sourced from the Federal Reserve Economic Database and the Michigan Consumer Survey. The survey measure of inflationary expectations is the mean response to the question, “What will be the percentage of price rise in the next 12 months?” We take the logarithm of real GDP, i.e., $100 \times \log(GDP_t)$. The quarterly urban CPI is transformed into the logarithm of annualized growth, i.e., $400 \times \log(\frac{CPI_t}{CPI_{t-1}})$.

5.2 Model Comparison Results

This section conducts a formal Bayesian model comparison of our proposed TVP-MCUC model against other competing models. The main objective of our model comparison exercise is to assess whether modeling features, such as a time-varying slope of the Phillips curve and correlated shocks, are favored by the data.

Table 1: List of competing models.

Model	Description
TVP-MCUC	Time-varying parameter (slope), multivariate correlated unobserved components (correlated shocks)
MCUC	Multivariate correlated unobserved components with constant slope
TVP-MUC	Time-varying slope with multivariate unobserved components (independent shocks)
MUC	Multivariate unobserved components (independent shocks) with constant slope

The first model we consider is our proposed TVP-MCUC model, specified in Equations (1) to (7). As discussed in Section 2, the TVP-MCUC extends the model proposed by Basistha and Nelson (2007) by allowing the slope of the Phillips curve to vary over time. The second model in the model comparison exercise, MCUC, proposed by Basistha and Nelson (2007), assumes a constant slope of the Phillips curve and correlated shocks. The third model we consider is the same as our proposed TVP-MCUC model, but with uncorrelated shocks. We refer to this model as TVP-MUC. Studies in the literature that use similar model specifications to TVP-MUC include Stella and Stock (2013), Chan et al. (2016), Berger et al. (2016), Fu (2020), and Kabundi et al. (2023). The last model is a conventional multivariate unobserved components model with a constant slope and uncorrelated shocks, which is referred to as MUC. Similar model specifications for MUC can be seen in Kuttner (1994); Roberts (2001). Since the TVP-MUC, MCUC, and MUC are just special cases of the TVP-MCUC model, the techniques described in Section 4, with slight modifications, can also be applied for computing the marginal likelihoods of these models. Table 1 summarizes the competing models in our model comparison exercise. The marginal likelihood for each model is computed based on 50,000 draws from the importance sampling density that is constructed using 10,000 posterior draws after a burn-in period of 5,000.

The log marginal likelihoods are reported in Table 2. A few findings can be drawn from these results. First, our results indicate that the proposed TVP-MCUC model performs best among all the models we consider. For example, the difference in log marginal likelihoods between our proposed TVP-MCUC and the second-best alternative model, TVP-MUC, is about 10, suggesting overwhelming evidence in favor of our proposed TVP-

MCUC. Second, it is evident that models specifying time-varying slopes of the Phillips curve perform significantly better than their counterparts with constant slopes. For example, for models with correlated shocks, the difference in the log marginal likelihoods between TVP-MCUC and MCUC is about 15, highlighting the empirical importance of extending the model of [Basistha and Nelson \(2007\)](#) by allowing for time-variation in the slopes of the Phillips curve. The importance of allowing for time-varying slopes can also be found from comparing the models with uncorrelated shocks, TVP-MUC and MUC. Third, models allowing for correlated shocks also substantially improve the model fit over their counterparts with uncorrelated shocks in terms of log marginal likelihood. This finding is supported by comparing the log marginal likelihoods of TVP-MCUC and MCUC with those of TVP-MUC and MUC.

In summary, the results of the model comparison demonstrate that jointly allowing for a time-varying slope and correlated shocks can lead to substantial statistical gains. These gains imply the importance of these two modeling features, suggesting that neglecting the time-varying slope and correlated shocks would miss important information for estimating the output gap using a MUC model. Moreover, these results corroborate the estimated output gap from our proposed model reported in the next section.

Table 2: Estimated log marginal likelihoods.

TVP-MCUC	MCUC	TVP-MUC	MUC
-849	-864	-858	-867

5.3 Estimates of Parameters

Table 3 reports the posterior means and the 90% credible intervals of the model parameters for TVP-MCUC. The estimated posterior means of the autoregressive coefficients ϕ_1 and ϕ_2 are 0.837 and 0.027, respectively, which indicates that the dynamics of the output gap are highly persistent. This result is consistent with the findings in [Basistha and Nelson \(2007\)](#). The posterior means of the coefficients associated with the inflation expectation, β_0 and β_1 , are -0.488 and 0.969 , respectively. This indicates that inflation

expectations are primarily driven by forward-looking components, as measured by survey expectations, while backward-looking inflation expectations also play a significant role.

To shed light on the importance of allowing for correlated shocks in our proposed TVP-MCUC, we present the posterior estimates for the correlation coefficients for the mixture inflation shock $\varepsilon_{\tau,t}$, the output gap shock $\varepsilon_{g,t}$ and the potential output shock $\varepsilon_{r,t}$ in Table 3. Specifically, we let $\rho_{\tau g}$ denote the correlation coefficient of $\varepsilon_{\tau,t}$ and $\varepsilon_{g,t}$, ρ_{gr} the correlation coefficient of $\varepsilon_{g,t}$ and $\varepsilon_{r,t}$, and $\rho_{\tau r}$ the correlation coefficient of $\varepsilon_{\tau,t}$ and $\varepsilon_{r,t}$. It is evident from our results that the 90% credible intervals associated with the posterior estimates for $\rho_{\tau g}$, ρ_{gr} and $\rho_{\tau r}$ all exclude 0, suggesting that accounting for correlations between shocks provides valuable information for estimating the output gap and improves model fit. A positive estimate of $\rho_{\tau g}$ and a negative estimate of ρ_{gr} are in line with the findings of Morley et al. (2003) and Basistha and Nelson (2007). This result can be interpreted as a “time-to-build effect”, wherein a significant positive permanent shock, such as technological advancement, causes the output gap to remain temporarily below its potential until it eventually converges (Morley et al., 2003). Similarly, a negative estimate of $\rho_{\tau r}$ suggests that a large negative inflation shock, such as an oil price collapse, leads to a temporary upward shift in potential output.

Table 3: Estimated posterior means and the 90% credible intervals for the model parameters of TVP-MCUC.

Parameter	Posterior mean	90% credible interval
<i>The autoregressive coefficients</i>		
ϕ_1	0.837	(0.687; 0.960)
ϕ_2	0.027	(-0.047; 0.206)
<i>The non-gap coefficients of the Phillips curve</i>		
β_0	-0.488	(-0.824; -0.129)
β_1	0.969	(0.913; 0.998)
<i>The variance of the innovations</i>		
σ_τ^2	0.947	(0.632; 1.310)
σ_r^2	2.620	(0.943; 6.215)
σ_g^2	2.523	(0.550; 6.701)
σ_δ^2	0.001	(0.001; 0.002)
<i>The correlation between the innovations</i>		
$\rho_{\tau g}$	0.817	(0.638; 0.930)
ρ_{gr}	-0.651	(-0.919; -0.310)
$\rho_{\tau r}$	-0.685	(-0.895; -0.389)

5.4 The Estimates of Output Gap

This section first presents the estimated output gap along with the corresponding potential output from our proposed TVP-MCUC model. We then compare the estimated output gap obtained from the TVP-MCUC model with those from the competing models TVP-MUC, MCUC, and MUC. Finally, we report the time-varying slope of the Phillips curve from the TVP-MCUC model to examine whether inflation can provide informative content for estimating the output gap.

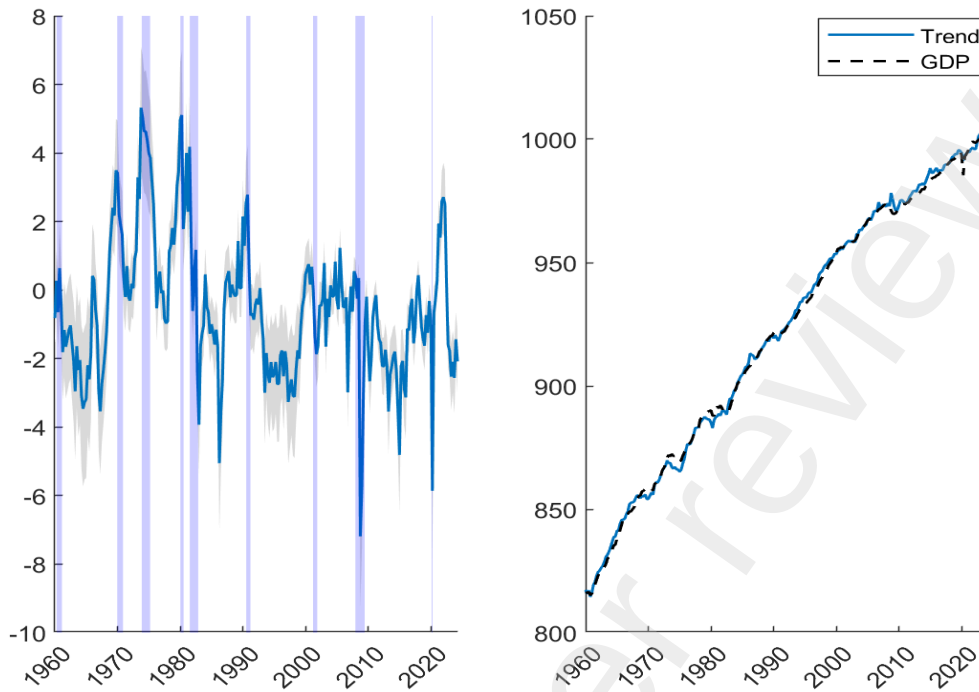


Figure 1: Estimates of output gap g_t and potential GDP r_t from the TVP-MCUC model. The left panel represents the output gap (solid blue line) and its 68% credible interval (shaded gray); shaded blue areas indicate the NBER recession periods. The right panel displays the potential output gap (solid blue line) and observed GDP (dashed black line).

Figure 1 displays the posterior means of the estimated output gap and potential output from our proposed TVP-MCUC model. The output gap estimates, presented in the left panel of Figure 1, have fluctuated with large magnitudes over the past four decades and have closely captured the NBER recession periods. It is also apparent that the estimated potential outputs, presented in the left panel of Figure 1, exhibit significant variations around the evolution of the real log GDP. This might be attributed to the significant correlations among the shocks to potential output, mixture inflation, and the output gap, as reported in Table 3.

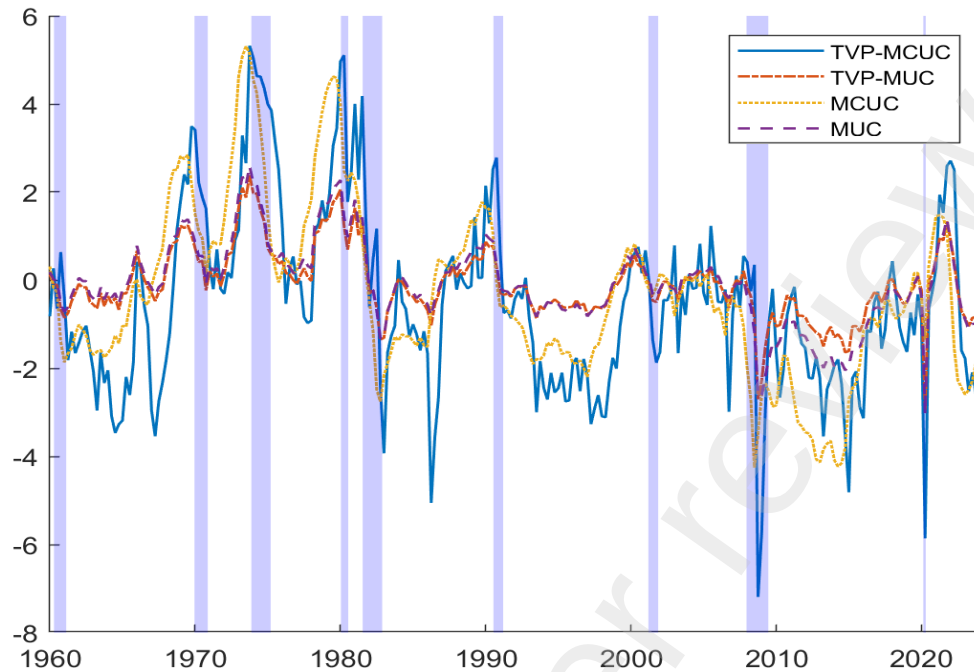


Figure 2: Comparison between the estimates of output gap g_t from the TVP-MCUC, TVP-MUC, MCUC and MUC models. The shaded blue areas indicate the NBER recession periods.

In Figure 2, we compare the output gap estimates from our proposed TVP-MCUC model with those of other competing models. Our results show that the output gap estimated by the TVP-MCUC model exhibits far greater variability than those of the other models. In particular, the TVP-MCUC estimates display a drastic drop during both the Great Recession of 2007–2009 and the COVID-19 pandemic. This suggests that ignoring the correlation among shocks and the time-varying slope of the Phillips curve within a unified multivariate unobserved components framework may lead to an underestimation of the variability in the output gap.

It is also interesting to see that, unlike other competing models, our proposed TVP-MCUC model identifies several substantial decreases in output gaps during periods not classified as NBER recessions, particularly in the 1980s. This finding might be due to the fact that negative inflation shocks—arising from both direct supply shocks and shifts in inflation expectations—may reduce production costs and boost total factor productivity

(TFP). As shown in Section 5.3, the significantly negative correlation between potential output shocks and these “mixture inflation shocks” implies that a decline in inflation raises the long-run path of output, leaving actual output trailing until it converges. This dynamic produces large negative output gaps during otherwise stable times. Historical oil price collapses illustrate this point. In 1986, for example, real GDP continued to grow steadily while potential output soared as oil prices plummeted to about 12 per barrel, returning to their 1974 levels. A similar pattern emerged in 2014, when oil prices fell by 44% in the first half of the year, again producing a substantial negative output gap.

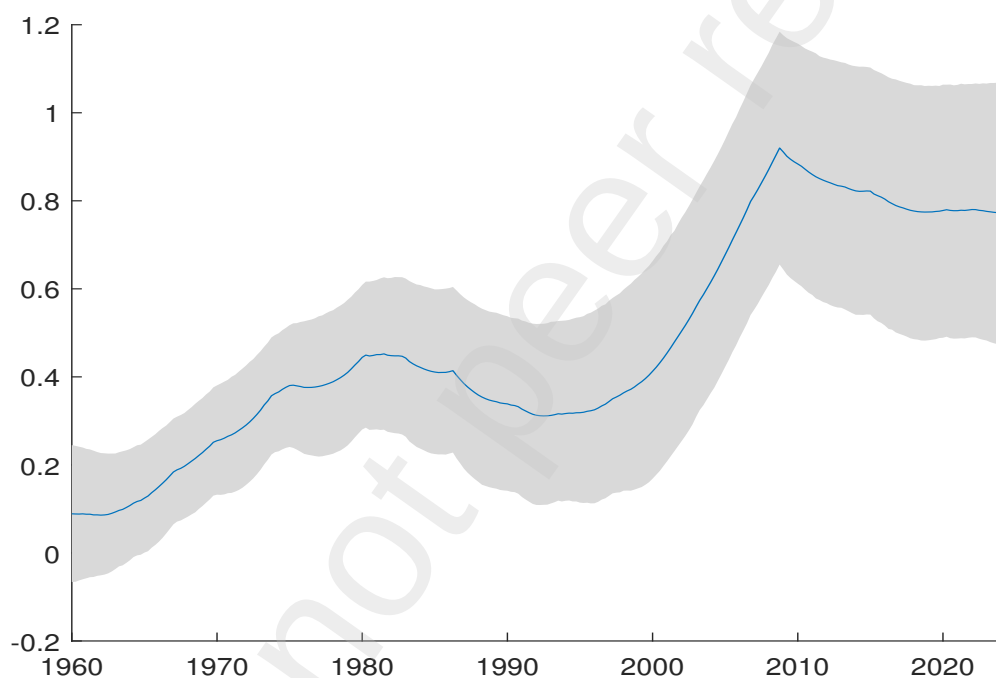


Figure 3: The estimated time-varying slope δ_t of the TVP-MCUC model. The shaded gray areas indicate the 68% credible interval.

Figure 3 plots the estimated slope of the Phillips curve from our proposed TVP-MCUC model. Our results reveal that the time-varying slope of the Phillips curve experiences significant rises and falls. It begins to decline in the early 1980s until the mid of 1990s, then rises to its peak during the Great Recession of 2007-2009, and then declines until the end of the sample period. The time-varying slope illustrates substantial time-variation. This finding shows that although there is a decline of the slope of the Phillips curve from

early 1980s to the mid of 1990s, overall the Phillips curve is well and alive, implying that inflation still play important role of estimating output gap.

6 Conclusion

In this paper, we propose a new mutivariate unobserved components model by jointly modeling correlated trend and cycle shocks and a time-varying Phillips curve slope within a unified multivariate unobserved components models framework. More specifically, we extend the multivariate correlated unobserved components model proposed by [Basistha and Nelson \(2007\)](#) to a TVP-MCUC model by incorporating a time-varying slope of the Phillips curve, achieving a better fit to inflation and output data despite the increased model complexity. Our TVP-MCUC model estimates a persistent output gap and reveals significantly larger negative output gaps during the Great Recession and the COVID-19 pandemic, suggesting that without jointly considering the time-varying relationship between inflation and the output gap and correlated shocks, multivariate unobserved components models may underestimate variability of output gap. Furthermore, the model captures substantial negative output gaps during stable economic periods. This occurs due to sudden upward shifts in the long-term output path resulting from large negative inflation shocks, leading actual output to temporarily fall below the new trend. Despite a decline in the Phillips curve slope from the early 1980s to the mid-1990s, we find that it remains robust, underscoring the importance of inflation in estimating the output gap.

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A MCMC sampler

In this section, we provide the details for implementing Step 3 – Step 5 of the posterior sampler outlined in Section 3.

Step 3. To sample $\boldsymbol{\theta}$, we first define

$$\begin{aligned}\mathbf{X}_{\theta,1} &= \begin{pmatrix} 1 & \pi_1^{se} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \\ \mathbf{X}_{\theta,2} &= \begin{pmatrix} 1 & \pi_2^{se} - \pi_2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & g_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \\ \mathbf{X}_{\theta,t} &= \begin{pmatrix} 1 & \pi_t^{se} - \pi_t & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & g_{t-1} & g_{t-2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \text{for } t = 3, \dots, T,\end{aligned}$$

$\boldsymbol{\mu}_{\theta,1} = (\delta_1 g_1, 0, 0, 0)'$ and $\boldsymbol{\mu}_{\theta,t} = (\pi_{t-1} + \delta_t g_t, 0, r_{t-1}, \delta_{t-1})'$ for $t = 2, \dots, T$. It can be shown that equation (17) can be rewritten as:

$$\mathbf{w} = \boldsymbol{\mu}_{\theta} + \mathbf{X}_{\theta} \boldsymbol{\theta} + \mathbf{u}, \quad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_T \otimes \boldsymbol{\Omega}),$$

where $\boldsymbol{\mu}_{\theta} = (\boldsymbol{\mu}'_{\theta,1}, \dots, \boldsymbol{\mu}'_{\theta,T})'$ and $\mathbf{X}_{\theta} = (\mathbf{X}'_{\theta,1}, \dots, \mathbf{X}'_{\theta,T})'$. The conditional posterior of $\boldsymbol{\theta}$ is a truncated Gaussian distribution:

$$(\boldsymbol{\theta} | \mathbf{r}, \boldsymbol{\delta}, \boldsymbol{\Sigma}, \sigma_{\delta}^2, \mathbf{y}) \sim \mathcal{TN}_{\{\beta_1 \in (0,1)\} \cap \{\phi \in \mathcal{R}\}}(\hat{\boldsymbol{\theta}}, \hat{\mathbf{K}}_{\theta}^{-1}),$$

where $\hat{\mathbf{K}}_{\theta} = \mathbf{X}'_{\theta} (\mathbf{I}_T \otimes \boldsymbol{\Omega}^{-1}) \mathbf{X}_{\theta} + \mathbf{V}_{\theta_0}^{-1}$ and $\hat{\boldsymbol{\theta}} = \hat{\mathbf{K}}_{\theta}^{-1} (\mathbf{X}'_{\theta} (\mathbf{I}_T \otimes \boldsymbol{\Omega}^{-1}) \mathbf{w} + \mathbf{V}_{\theta_0}^{-1} \boldsymbol{\theta}_0)$ with $\boldsymbol{\theta}_0 = (\beta_{00}, \beta_{10}, \phi_0, \mu_0, r_{00}, \delta_{00})'$ and $\mathbf{V}_{\theta_0} = \text{diag}(V_{\beta_{00}}, V_{\beta_{10}}, \mathbf{V}_{\phi_0}, V_{\mu_0}, V_{r_0}, V_{\delta_0})$. Hence, draws

of $\boldsymbol{\theta}$ can be obtained by using an accept-reject algorithm.

Step 4. Let $\boldsymbol{\varepsilon} = \mathbf{A}\tilde{\mathbf{x}} - \mathbf{c}$ and write $\boldsymbol{\varepsilon} = (\boldsymbol{\varepsilon}_1', \dots, \boldsymbol{\varepsilon}_T')'$. By combining the likelihood defined by equation (13) and the inverse-Wishart prior for $\boldsymbol{\Sigma}$, it is easy to show that the conditional posterior of $\boldsymbol{\Sigma}$ follows an inverse-Wishart distribution:

$$(\boldsymbol{\Sigma}|\mathbf{r}, \boldsymbol{\delta}, \boldsymbol{\theta}, \sigma_\delta^2, \mathbf{y}) \sim \mathcal{IW}(\mathbf{S} + \sum_{t=1}^T \boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t', \nu + T).$$

Step 5. Given the inverse-Gamma prior on σ_δ^2 , the conditional posterior distribution of σ_δ^2 is also an inverse-Gamma distribution. That is,

$$(\sigma_\delta^2|\boldsymbol{\delta}, \mathbf{y}) \sim \mathcal{IG}\left(\alpha + \frac{T}{2}, \gamma + \frac{1}{2} \sum_{t=1}^T (\delta_t - \delta_{t-1})^2\right).$$

B Integrated Likelihood and Implied Prior

This section provides the analytical expression for the integrated likelihood $p(\mathbf{y}|\boldsymbol{\eta}_*)$ and the implied prior $p(\boldsymbol{\eta}_*)$. Recall that $\mathbf{y} = (\boldsymbol{\pi}', \mathbf{z}')'$. An easy way to obtain the integrated likelihood $p(\mathbf{y}|\boldsymbol{\eta}_*) = p(\mathbf{y}|\boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma})$ is to notice that $(\mathbf{r}', \mathbf{y}'|\boldsymbol{\delta}, \boldsymbol{\theta})$ is Gaussian distributed, which is given in (14). Using the standard property of Gaussian distribution, it implies that the marginal distribution of $(\mathbf{y}'|\boldsymbol{\delta}, \boldsymbol{\theta}, \sigma_\delta^2)$ is also Gaussian. Specifically, from the result given in (14), we have

$$(\mathbf{y}|\boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma}) \sim \mathcal{N}(\boldsymbol{\mu}_y, \mathbf{D}_y^{-1}),$$

where $\mathbf{D}_y = \mathbf{K}_y - \mathbf{K}_{r,y}' \mathbf{K}_r^{-1} \mathbf{K}_{r,y}$. Therefore, the integrated likelihood $p(\mathbf{y}|\boldsymbol{\eta}_*)$ is given by

$$p(\mathbf{y}|\boldsymbol{\eta}_*) = p(\mathbf{y}|\boldsymbol{\delta}, \boldsymbol{\theta}, \boldsymbol{\Sigma}) = (2\pi)^{-T} |\mathbf{D}_y|^{\frac{1}{2}} e^{-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu}_y)' \mathbf{K}_y (\mathbf{y} - \boldsymbol{\mu}_y)}.$$

Next, we will derive implied prior $p(\boldsymbol{\eta}_*) = p(\boldsymbol{\delta}|\delta_0)p(\boldsymbol{\Sigma})p(\boldsymbol{\theta})$. Given the specification (2)

and the prior for σ_δ^2 given in (8), we have

$$p(\boldsymbol{\delta}|\delta_0, \sigma_\delta^2) = (2\pi)^{-\frac{T}{2}} (\sigma_\delta^2)^{-\frac{T}{2}} e^{-\frac{1}{2\sigma_\delta^2} \sum_{t=1}^T (\delta_t - \delta_{t-1})^2}, \quad p(\sigma_\delta^2) = \frac{\gamma^\alpha}{\Gamma(\alpha)} (\sigma_\delta^2)^{-\alpha-1} e^{-\frac{\gamma}{\sigma_\delta^2}}.$$

Thus the implied prior for $\boldsymbol{\delta}$ given δ_0 is given by

$$\begin{aligned} p(\boldsymbol{\delta}|\delta_0) &= \int p(\boldsymbol{\delta}|\delta_0, \sigma_\delta^2) p(\sigma_\delta^2) d\sigma_\delta^2 \\ &= (2\pi)^{-\frac{T}{2}} \frac{\gamma^\alpha}{\Gamma(\alpha)} \int (\sigma_\delta^2)^{-(\alpha+\frac{T}{2})-1} e^{-(\gamma+\frac{1}{2} \sum_{t=1}^T (\delta_t - \delta_{t-1})^2)/\sigma_\delta^2} d\sigma_\delta^2 \\ &= (2\pi)^{-\frac{T}{2}} \frac{\Gamma(\hat{\alpha})}{\Gamma(\alpha)} \frac{\gamma^\alpha}{\hat{\gamma}^{\hat{\alpha}}}, \end{aligned} \quad (28)$$

where $\hat{\alpha} = \alpha + \frac{T}{2}$ and $\hat{\gamma} = \gamma + \frac{1}{2} \sum_{t=1}^T (\delta_t - \delta_{t-1})^2$. The prior for $\boldsymbol{\theta}$ is given in (8), that implies

$$p(\boldsymbol{\theta}) = \frac{c_p}{\Phi\left(\frac{1-\beta_{00}}{\sqrt{V_{\beta_{00}}}}\right) - \Phi\left(\frac{-\beta_{00}}{\sqrt{V_{\beta_{00}}}}\right)} (2\pi)^{-\frac{7}{2}} |\mathbf{V}_{\theta_0}|^{-\frac{1}{2}} e^{-\frac{1}{2}(\boldsymbol{\theta}-\boldsymbol{\theta}_0)'\mathbf{V}_{\theta_0}^{-1}(\boldsymbol{\theta}-\boldsymbol{\theta}_0)} \mathbf{1}(\boldsymbol{\phi} \in \mathcal{R}, 0 < \beta_1 < 1), \quad (29)$$

where $c_p = \int_{\mathcal{R}} (2\pi)^{-1} |\mathbf{V}_{\phi_0}|^{-\frac{1}{2}} e^{-\frac{1}{2}(\boldsymbol{\phi}-\boldsymbol{\phi}_0)'\mathbf{V}_{\phi_0}^{-1}(\boldsymbol{\phi}-\boldsymbol{\phi}_0)} d\boldsymbol{\phi}$ is the unknown normalizaing constant of the prior for $\boldsymbol{\phi}$, $\Phi(\cdot)$ is the cumulative distribution of standard Gaussian distribution, $\Gamma_3(\cdot)$ is the multivariate gamma function, and $\boldsymbol{\theta}_0$ and $\mathbf{V}_{\theta_0}$ are defined in Appendix A.

Finally, given the Inverse-Wishart prior for $\boldsymbol{\Sigma}$ specified in (8), and the results in (28) and (29), we have

$$\begin{aligned} p(\boldsymbol{\eta}_*) &= p(\boldsymbol{\delta}|\delta_0) p(\boldsymbol{\Sigma}) p(\boldsymbol{\theta}), \\ &= c_p \tilde{p}(\boldsymbol{\eta}_*), \end{aligned}$$

where

$$\tilde{p}(\boldsymbol{\eta}_*) = \frac{(2\pi)^{-\frac{T+7}{2}} 2^{-\frac{3\nu}{2}} |\mathbf{S}|^{\frac{\nu}{2}} \gamma^\alpha \Gamma(\hat{\alpha})}{|\boldsymbol{\Sigma}|^{-\frac{\nu+4}{2}} |\mathbf{V}_{\theta_0}|^{-\frac{1}{2}} \Gamma_3(\frac{\nu}{2}) \hat{\gamma}^{\hat{\alpha}} \Gamma(\alpha)} e^{-\frac{1}{2}((\boldsymbol{\theta}-\boldsymbol{\theta}_0)'\mathbf{V}_{\theta_0}^{-1}(\boldsymbol{\theta}-\boldsymbol{\theta}_0) + \text{tr}(\mathbf{S}\boldsymbol{\Sigma}^{-1}))} \frac{\mathbf{1}(\boldsymbol{\phi} \in \mathcal{R}, 0 < \beta_1 < 1)}{\Phi\left(\frac{1-\beta_{10}}{\sqrt{V_{\beta_{10}}}}\right) - \Phi\left(\frac{-\beta_{10}}{\sqrt{V_{\beta_{10}}}}\right)}.$$

C Importance Sampling Densities

Recall that we consider the importance sampling density function $q(\boldsymbol{\eta}_*; \hat{\mathbf{v}}) = q_{\bar{\delta}}(\bar{\boldsymbol{\delta}}; \hat{\mathbf{v}}_{\bar{\delta}}) \times q_{\bar{\theta}}(\bar{\boldsymbol{\theta}}; \hat{\mathbf{v}}_{\bar{\theta}}) \times q_{\Sigma}(\boldsymbol{\Sigma}; \hat{\mathbf{v}}_{\Sigma})$, where $\bar{\boldsymbol{\delta}} = (\delta_0, \boldsymbol{\delta}')'$, $\bar{\boldsymbol{\theta}} = (\beta_0, \beta_1, \boldsymbol{\phi}', \mu, r_0)'$ and

$$\begin{aligned} q_{\bar{\delta}}(\bar{\boldsymbol{\delta}}; \hat{\mathbf{v}}_{\bar{\delta}}) &= f_{\mathcal{N}}(\bar{\boldsymbol{\delta}}; \hat{\boldsymbol{\psi}}_{\bar{\delta}}, \hat{\mathbf{D}}_{\bar{\delta}}), \\ q_{\Sigma}(\boldsymbol{\Sigma}; \hat{\mathbf{v}}_{\Sigma}) &= f_{\mathcal{IW}}(\boldsymbol{\Sigma}; \hat{\mathbf{S}}, \hat{\nu}), \\ q_{\bar{\theta}}(\bar{\boldsymbol{\theta}}; \hat{\mathbf{v}}_{\bar{\theta}}) &= f_{\mathcal{TN}_{\{\beta_1 \in (0,1)\} \cap \{\phi \in \mathcal{R}\}}}(\bar{\boldsymbol{\theta}}; \hat{\boldsymbol{\psi}}_{\bar{\theta}}, \hat{\mathbf{D}}_{\bar{\theta}}). \end{aligned}$$

Here $f_{\mathcal{N}}$ and $f_{\mathcal{IW}}$ denoting Gaussian and inverse-Wishart densities, respectively, $f_{\mathcal{TN}_A}$ denotes the truncated Gaussian density with its support restricted on the region A , and we denote $\hat{\mathbf{v}}_{\bar{\delta}} = (\hat{\boldsymbol{\psi}}_{\bar{\delta}}, \hat{\mathbf{D}}_{\bar{\delta}})$, $\hat{\mathbf{v}}_{\bar{\theta}} = (\hat{\boldsymbol{\psi}}_{\bar{\theta}}, \hat{\mathbf{D}}_{\bar{\theta}})$ and $\hat{\mathbf{v}}_{\Sigma} = (\hat{\mathbf{S}}, \hat{\nu})$. In what follows, we describe how to find the maximizer $\hat{\mathbf{v}} = (\hat{\mathbf{v}}_{\bar{\delta}}, \hat{\mathbf{v}}_{\bar{\theta}}, \hat{\mathbf{v}}_{\Sigma})$ using the posterior draws $(\bar{\boldsymbol{\delta}}^{(m)}, \bar{\boldsymbol{\theta}}^{(m)}, \boldsymbol{\Sigma}^{(m)})$ for $m = 1, \dots, M$.

Since $\mathbf{v}_{\bar{\delta}}$ is high-dimensional, solving the maximization problem (23) using only the information from the posterior draws $(\bar{\boldsymbol{\delta}}^{(1)}, \dots, \bar{\boldsymbol{\delta}}^{(M)})$ might lead to poor estimation accuracy. Hence, we adopt the approach of Chan (2023) by introducing an auxiliary model to reduce the number of parameters that need to be estimated. Specifically, we consider the following auxiliary model for $\bar{\boldsymbol{\delta}} = (\delta_0, \boldsymbol{\delta}')'$:

$$\delta_t = a_t + \rho \delta_{t-1} + e_{\delta,t}, \quad e_{\delta,t} \sim \mathcal{N}(0, \xi_t), \quad (30)$$

for $t = 1, \dots, T$ and $\delta_0 \sim \mathcal{N}(a_0, \xi_0)$. We can represent this auxiliary model more compactly as

$$\mathbf{H}_{\rho} \bar{\boldsymbol{\delta}} = \mathbf{a} + \mathbf{e}_{\delta}, \quad \mathbf{e}_{\delta} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Xi}),$$

where $\mathbf{a} = (a_0, \dots, a_T)'$, $\Xi = \text{diag}(\boldsymbol{\xi})$ with $\boldsymbol{\xi} = (\xi_0, \dots, \xi_T)'$, and

$$\mathbf{H}_\rho = \begin{pmatrix} 1 & 0 & \cdots & \cdots & 0 \\ -\rho & 1 & \ddots & \ddots & \vdots \\ 0 & -\rho & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & -\rho & 1 \end{pmatrix}.$$

This implies that $\bar{\boldsymbol{\delta}} \sim \mathcal{N}(\mathbf{H}_\rho^{-1}\mathbf{a}, (\mathbf{H}_\rho'\Xi^{-1}\mathbf{H}_\rho)^{-1})$. Our goal is to maximize the following log-likelihood of this auxiliary model with respect to $(\rho, \mathbf{a}, \boldsymbol{\xi})$ given the posterior draws $(\bar{\boldsymbol{\delta}}^{(1)}, \dots, \bar{\boldsymbol{\delta}}^{(M)})$:

$$l_{\bar{\boldsymbol{\delta}}}(\rho, \mathbf{a}, \boldsymbol{\xi}) = \frac{(T+1)M}{2} \log(2\pi) - \frac{M}{2} \log |\Xi| - \frac{1}{2} \sum_{m=1}^M \left(\mathbf{H}_\rho \bar{\boldsymbol{\delta}}^{(m)} - \mathbf{a} \right)' \Xi^{-1} \left(\mathbf{H}_\rho \bar{\boldsymbol{\delta}}^{(m)} - \mathbf{a} \right).$$

Given any ρ , it can be shown that the analytical solution $\hat{\mathbf{a}}(\rho)$ and $\hat{\boldsymbol{\xi}}(\rho)$ are functions of ρ , which are given by

$$\hat{\mathbf{a}}(\rho) = \frac{1}{M} \sum_{m=1}^M \mathbf{H}_\rho \bar{\boldsymbol{\delta}}^{(m)}, \quad \hat{\boldsymbol{\xi}}(\rho) = \frac{1}{R} \sum_{m=1}^M (\mathbf{H}_\rho \bar{\boldsymbol{\delta}}^{(m)} - \hat{\mathbf{a}}(\rho))^2, \quad (31)$$

Substituting the above expressions into the log-likelihood function, we can obtain a one-dimensional concentrated log-likelihood $l_{\bar{\boldsymbol{\delta}}}(\rho, \hat{\mathbf{a}}(\rho), \hat{\boldsymbol{\xi}}(\rho))$ that depends only on ρ . The solution $\hat{\rho}$ can be obtained by maximizing $l_{\bar{\boldsymbol{\delta}}}(\rho, \hat{\mathbf{a}}(\rho), \hat{\boldsymbol{\xi}}(\rho))$ with respect to ρ numerically. Substituting $\hat{\rho}$ into the expressions in (31), we can obtain the maximizer $\hat{\mathbf{a}} = \hat{\mathbf{a}}(\hat{\rho})$ and $\hat{\boldsymbol{\xi}} = \hat{\boldsymbol{\xi}}(\hat{\rho})$. Therefore the mean $\hat{\boldsymbol{\psi}}_{\bar{\boldsymbol{\delta}}}$ and the covariance matrix $\hat{\mathbf{D}}_{\bar{\boldsymbol{\delta}}}$ for the importance sampling distribution are respectively set to $\hat{\boldsymbol{\psi}}_{\bar{\boldsymbol{\delta}}} = \mathbf{H}_{\hat{\rho}}^{-1}\hat{\mathbf{a}}$ and $\hat{\mathbf{D}}_{\bar{\boldsymbol{\delta}}} = (\mathbf{H}_{\hat{\rho}}'\hat{\Xi}^{-1}\mathbf{H}_{\hat{\rho}})^{-1}$ with $\hat{\Xi} = \text{diag}(\hat{\boldsymbol{\xi}})$.

For the maximization problem (24), its solution $\hat{\mathbf{v}}_{\Sigma} = (\hat{\mathbf{S}}, \hat{\nu})$ is the maximizer of the following log-likelihood:

$$l_{\Sigma}(\mathbf{S}, \nu) = -\frac{\nu M}{2} (3 \log(2) - \log |\mathbf{S}|) - M \log \Gamma_3\left(\frac{\nu}{2}\right) - \frac{\nu+4}{2} \sum_{m=1}^M \log |\Sigma^{(m)}| - \frac{1}{2} \sum_{m=1}^M \text{tr}(\mathbf{S}(\Sigma^{(m)})^{-1}).$$

It can be shown that, for any ν , we can find the analytical solution $\widehat{\mathbf{S}}(\nu)$ as a function of ν , which is given by

$$\widehat{\mathbf{S}}(\nu) = \frac{\nu M}{\sum_{m=1}^M (\boldsymbol{\Sigma}^{(m)})^{-1}}. \quad (32)$$

Now substituting $\widehat{\mathbf{S}}(\nu)$ into the log-likelihood function gives the concentrated log-likelihood $l_{\Sigma}(\widehat{\mathbf{S}}(\nu), \nu)$ that depends only on ν . Hence we can first find the solution $\widehat{\nu}$ numerically, and then obtain the maximizer $\widehat{\mathbf{S}} = \widehat{\mathbf{S}}(\widehat{\nu})$.

To find maximizer $\widehat{\mathbf{v}}_{\bar{\theta}} = (\widehat{\boldsymbol{\psi}}_{\bar{\theta}}, \widehat{\mathbf{D}}_{\bar{\theta}})$ for the maximization problem (25), we further assume $\widehat{\mathbf{D}}_{\bar{\theta}}$ to be a block diagonal matrix. Specifically, we write $\widehat{\mathbf{D}}_{\bar{\theta}} = \text{diag}(\widehat{D}_{\beta_0}, \widehat{D}_{\beta_1}, \widehat{\mathbf{D}}_{\phi}, \widehat{D}_{\mu}, \widehat{D}_{r_0})$ where $\widehat{D}_i > 0$ for $i \in \{\beta_0, \beta_1, \mu, r_0\}$ and $\widehat{\mathbf{D}}_{\phi}$ is a 2×2 positive definite matrix. We write $\widehat{\boldsymbol{\psi}}_{\bar{\theta}} = (\widehat{\psi}_{\beta_0}, \widehat{\psi}_{\beta_1}, \widehat{\boldsymbol{\psi}}'_{\phi}, \widehat{\psi}_{\mu}, \widehat{\psi}_{r_0})'$ with $\widehat{\mu}_i$ being a scalar for $i \in \{\beta_0, \beta_1, \mu, r_0\}$ and $\widehat{\boldsymbol{\psi}}_{\phi}$ a 2×1 vector. With this setting, the truncated Gaussian importance sampling density $q_{\bar{\theta}}(\bar{\boldsymbol{\theta}}; \widehat{\mathbf{v}}_{\bar{\theta}}) = f_{\mathcal{TN}_{\{\beta_1 \in (0,1)\} \cap \{\phi \in \mathcal{R}\}}}(\bar{\boldsymbol{\theta}}; \widehat{\boldsymbol{\psi}}_{\bar{\theta}}, \widehat{\mathbf{D}}_{\bar{\theta}})$ takes the form

$$f_{\mathcal{N}}(\beta_0; \widehat{\psi}_{\beta_0}, \widehat{D}_{\beta_0}) f_{\mathcal{TN}_{\{\beta_1 \in (0,1)\}}}(\beta_1; \widehat{\psi}_{\beta_0}, \widehat{D}_{\beta_0}) f_{\mathcal{TN}_{\{\phi \in \mathcal{R}\}}}(\boldsymbol{\phi}; \widehat{\boldsymbol{\psi}}_{\phi}, \widehat{\mathbf{D}}_{\phi}) f_{\mathcal{N}}(\mu; \widehat{\psi}_{\mu}, \widehat{D}_{\mu}) f_{\mathcal{N}}(r_0; \widehat{\psi}_{r_0}, \widehat{D}_{r_0}).$$

Therefore, given the posterior draws $(\beta_0^{(m)}, \beta_1^{(m)}, \boldsymbol{\phi}^{(m)}, \mu^{(m)}, r_0^{(m)})$ for $m = 1, \dots, M$, and using the standard results of the maximum likelihood estimation, we have the following solution:

$$\begin{aligned} \widehat{\psi}_{\beta_0} &= \frac{1}{M} \sum_{m=1}^M \beta_0^{(m)}, & \widehat{D}_{\beta_0} &= \frac{1}{M} \sum_{m=1}^M (\beta_0^{(m)} - \widehat{\psi}_{\beta_0})^2, \\ \widehat{\psi}_{\beta_1} &= \frac{1}{M} \sum_{m=1}^M \beta_1^{(m)}, & \widehat{D}_{\beta_1} &= \frac{1}{M} \sum_{m=1}^M (\beta_1^{(m)} - \widehat{\psi}_{\beta_1})^2, \\ \widehat{\boldsymbol{\psi}}_{\phi} &= \frac{1}{M} \sum_{m=1}^M \boldsymbol{\phi}^{(m)}, & \widehat{\mathbf{D}}_{\phi} &= \frac{1}{M} \sum_{m=1}^M (\boldsymbol{\phi}^{(m)} - \widehat{\boldsymbol{\psi}}_{\phi})(\boldsymbol{\phi}^{(m)} - \widehat{\boldsymbol{\psi}}_{\phi})', \\ \widehat{\psi}_{\mu} &= \frac{1}{M} \sum_{m=1}^M \mu^{(m)}, & \widehat{D}_{\mu} &= \frac{1}{M} \sum_{m=1}^M (\mu^{(m)} - \widehat{\psi}_{\mu})^2, \\ \widehat{\psi}_{r_0} &= \frac{1}{M} \sum_{m=1}^M r_0^{(m)}, & \widehat{D}_{r_0} &= \frac{1}{M} \sum_{m=1}^M (r_0^{(m)} - \widehat{\psi}_{r_0})^2. \end{aligned}$$

Finally, with the maximizer $\hat{\mathbf{v}} = (\hat{\boldsymbol{\psi}}_{\bar{\delta}}, \hat{\mathbf{D}}_{\bar{\delta}}, \hat{\boldsymbol{\psi}}_{\bar{\theta}}, \hat{\mathbf{D}}_{\bar{\theta}}, \hat{\mathbf{S}}, \hat{\nu})$, the importance sampling density is given by

$$q(\boldsymbol{\eta}_*; \hat{\mathbf{v}}) = c_q \tilde{q}(\boldsymbol{\eta}_*; \hat{\mathbf{v}}),$$

where c_q is an unknown normalizing constant, which has the following expression:

$$c_q = \int_{\mathcal{R}} (2\pi)^{-1} |\hat{\mathbf{D}}_{\phi}|^{-\frac{1}{2}} e^{-\frac{1}{2}(\phi - \hat{\boldsymbol{\psi}}_{\phi})' \hat{\mathbf{D}}_{\phi}^{-1} (\phi - \hat{\boldsymbol{\psi}}_{\phi})} d\phi$$

and

$$\tilde{q}(\boldsymbol{\eta}_*; \hat{\mathbf{v}}) = \frac{(2\pi)^{-\frac{T+1}{2}} 2^{-\frac{3\hat{\nu}}{2}} |\hat{\mathbf{S}}|^{\frac{\hat{\nu}}{2}}}{|\hat{\mathbf{D}}_{\bar{\delta}}|^{\frac{1}{2}} |\hat{\mathbf{D}}_{\bar{\theta}}|^{\frac{1}{2}} |\boldsymbol{\Sigma}|^{\frac{\hat{\nu}+4}{2}} \Gamma_3\left(\frac{\hat{\nu}}{2}\right)} e^{-\frac{1}{2}((\bar{\delta} - \hat{\boldsymbol{\psi}}_{\bar{\delta}})' \hat{\mathbf{D}}_{\bar{\delta}}^{-1} (\bar{\delta} - \hat{\boldsymbol{\psi}}_{\bar{\delta}}) + (\bar{\boldsymbol{\theta}} - \hat{\boldsymbol{\psi}}_{\bar{\theta}})' \hat{\mathbf{D}}_{\bar{\theta}}^{-1} (\bar{\boldsymbol{\theta}} - \hat{\boldsymbol{\psi}}_{\bar{\theta}}) + \text{tr}(\hat{\mathbf{S}} \boldsymbol{\Sigma}^{-1}))} \frac{\mathbf{1}(\phi \in \mathcal{R}, 0 < \beta_1 < 1)}{\Phi\left(\frac{1 - \hat{\psi}_{\beta_1}}{\sqrt{\hat{D}_{\beta_1}}}\right) - \Phi\left(\frac{-\hat{\psi}_{\beta_1}}{\sqrt{\hat{D}_{\beta_1}}}\right)}.$$